

Embedded Engineer

Dileep

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Summary:

- Around 5 years of experience as an embedded software developer, in design, development, verification and debugging in the field of automotive Embedded and informative systems.
- Experience with C, C++, JAVA, Embedded C and QNX/Linux expertise with Real time Embedded Projects and mixed-language programming techniques.
- Developed tailored HAL modules for TV OS stacks, specializing in MTK 5583, 5691, and NVT72690 chipsets.
- Designed and implemented test plans and strategies for embedded systems, including firmware testing and hardware-software integration testing, meeting project requirements and deadlines.
- Skilled in CLI tool development for efficient control and configuration of HAL services.
- Proficient in implementing Build Root configuration and QEMU testing procedures, optimizing cross-compilation and emulation for real-time Linux kernel.
- Proficient in kernel configuration, device drivers' development, and kernel module management.
- Familiarity with kernel internals, memory management, and file systems.
- Designing and implementing automated test scripts using the Python programming language to replace manual testing processes.
- Dynamic and results-oriented professional with extensive experience in test design and script creation for Hardware-in-Loop (HIL) testing.
- Seasoned Electrical Engineer with extensive experience in designing, developing, and optimizing High Voltage (HV) systems and Battery Management Systems (BMS) for electric vehicles.
- Skilled in implementing Auto Sar Application software layer for embedded systems, ensuring compliance with automotive safety standards, and streamlining software development processes.
- Utilized Git and SVN extensively for version control, enabling efficient collaboration, code review, and branching strategies within distributed development teams.
- Utilized SVN for version control, Make and CMake for build automation, and Docker for containerization, streamlining development and deployment processes.
- Sound knowledge on embedded communication protocols CAN, LIN and FLEXRAY
- Experienced in writing application firmware for microcontrollers, incorporating CAN/J1939 and TCP/IP protocols, with a focus on delivering efficient and optimized code.
- Experience in development of device driver for USART, SPI, I2C, J1939, CAN and interfacing memory chips like FLASH, SRAM and SDRAM.
- Utilized issue tracking tools such as JIRA and TestRail to manage test cases, track defects, and report testing progress to project managers and clients.
- Expert in Testing and Debugging the Embedded C code and handling various Interrupts in various environments and good in Assembly languages for 8051, ARM, MSP430, PIC Core.
- Experienced in Hardware and Software design of microcontroller embedded systems.
- Hands on experience in Configuration and AUTOSAR System testing of ECUs.
- Enhanced the VxWorks network stack in support of the communications infrastructure.
- Strong Experience with Linux/UNIX, RTOS with 32-bit microprocessor architecture.
- Good knowledge of HIL testing and vehicle diagnostics using CANape, CANoe, Lab Car, and Luna Win and Mat lab.
- Leveraged debugging tools including GDB, Valgrind, Strace, and Ptrace for troubleshooting and optimizing code performance, ensuring robustness and reliability of software solutions.
- Actively participated in Agile-based project workflows, including Scrum or Kanban, facilitating iterative development cycles, and enabling adaptability to changing project requirements.

- Participated actively in team meetings, brainstorming sessions, and code reviews to foster collaboration and drive innovation.

TECHNICAL SKILLS:

Operating Systems	VxWorks, Windows, Unix/Linux, RTOS, Mac OS etc
Programming Languages	Embedded C/ C++, VHDL, Assembly and Python, JAVA
Processor	8051, AVR, PIC Microcontrollers, ARM Cortex M3, ARM Embedded Board
Tools	SVN,Make,CMake,docker,vim,Microsoft studio,Eclipse,GDB,valgrind,strace,postrace,HAT, HCA,NI LabVIEW 2012, MATLAB, Electronic Workbench (MultiSim, Commsim and Ultiboard), BMS,KIEL, ModelSim, SimuTAG, Vector CAN analyzer, CANoe,UDS,DoCAN etc., wire shark,GIT.
Communication Protocols	UART, I ² C, I2S, CAN, SPI, RS232, PWM, PID Controllers,CAN/J1939,TCP/IP.
Data bases	Oracle, PL/SQL

Client: Vizio

August 2023 to till date

Role: Embedded Software engineer/C++ developer

Responsibilities:

- Developed HAL modules for TV OS stack on Chipsets MTK 5583,5691 and NVT72690 (Input Manager, Audio Quality, Picture Quality, HDMI Manager).
- Created a CLI tool for streamlined control and configuration of HAL service, enhancing user accessibility.
- Engineered a user-friendly CLI tool with advanced testing and debugging, improving HAL service reliability and stability.
- Implemented video playback functionality with diverse codec and streaming protocol support for Vizio TV users.
- Developed and maintained operating system drivers for embedded platforms, ensuring seamless integration with the kernel and Linux-based environments.
- Collaborated with cross-functional teams to diagnose and resolve kernel-level issues, ensuring system stability.
- Involved in writing optimization techniques performance tuning in Oracle SQL, PL/SQL.
- Integrated third-party libraries and APIs to enhance Vizio TV features like streaming services and smart home integration.
- Developed customized distro image using Build Root for efficient software development in embedded systems.
- Implemented Build Root configuration and QEMU testing, optimizing the development process for embedded solutions.
- Documented test cases, test procedures, and test results to ensure traceability and facilitate knowledge transfer within the project team.
- Effectively resolved blocker, critical, and major issues by employing strategic prioritization and collaborative problem-solving, ensuring project continuity and system reliability.
- Tested Airplay video certification test cases using HAT and HCA Tool.
- Managed dependencies, libraries, and configurations within the Vizio project environment, facilitating seamless integration and deployment of software solutions.

Environment: ARM Embedded Board, C, C++, RTOS, Ubuntu, Python, Shell Script, JSON, Make, CMake, AWS CLOUD, BSP, GDB, SVN, Visual studio, PL/SQL, FileZilla, netstat, sysstat, init services, system services, RPM Build tools, GIT, Jenkins, Docker, windows, Linux.

Client: Jing-Jin Electric North America, Farmington MI

Sept 2022 to Aug 2023

Role: Embedded Developer

Responsibilities:

- Developed, tested, and maintained software for proprietary CAN-based vehicle diagnostic protocol (V850).
- Developed embedded software for trailer brake controllers (braking functionality, diagnostics, CAN communication).
- Led the design and development of High Voltage (HV) systems for electric vehicles, ensuring compliance with safety standards and regulations.
- Spearheaded the integration and optimization of Battery Management Systems (BMS) to enhance performance and reliability in electric vehicle applications.
- Implemented Auto Sar Application software layer for embedded systems, streamlining software development processes and ensuring compliance with automotive safety standards (ISO 26262).
- Prepared functional specifications using MATLAB, Simu link, and State flow.
- Developed and maintained PC tools and build scripts.
- Worked on software such as the CANoe for development & analysis of the Car ECU network and CANalyzer for configuring the bus communication, analyzing the report logs etc., USB, I2C, My info etc.
- Created SQL queries, Functions for the Database layer by studying the required business objects and validating them with Stored Procedures
- Implemented diagnostics protocols such as UDS, DoCAN, and DoIP, facilitating efficient vehicle diagnostics and maintenance processes.
- Experience with continuous integration and continuous deployment (CI/CD) practices, contributing to a seamless software delivery pipeline and ensuring rapid and reliable software releases.
- Perform integration of new system software components using Embedded C/C++, JAVA and QNX/Linux expertise programming languages.
- Involved in designing and modifying drivers for low-level communication protocol like CAN, LIN, SPI and I2C.
- Successfully directing the development efforts for implementing the projects and designing the interface between the LCD, Seven Segments, Motors, and sensors with 8051, AVR and ARM7 microcontroller.
- Participated in team meetings and discussions to review project requirements, identify testing priorities, and plan testing activities, contributing to project success.

Environment: Embedded C/C++, JAVA, CANoe, CAN Analyzer, Matlab, Autosar standards and guidelines., Linux, Oracle, openssl, SQL, Shell, POSIX Threads, Visual Studio, Eclipse, Git, gdb, REST APIs, AWS cloud services, PL/SQL, JIRA

Client: Joyson - San Antonio, TX

January 2022 to Aug 2022

Role: Embedded Software Engineer

Responsibilities:

- Designed, debugged, and modified drivers for low-level communication protocols such as CAN, LIN, SPI, I2C, TCP/IP, Modbus, and CAN/J1939.
- Performed failsafe tests on various ECUs and verified whether ABS and TC software subsystems react appropriately in the simulated environment, including CAN/J1939 and TCP/IP communication.

- Demonstrated understanding of AUTOSAR specifications for SPI (Serial Peripheral Interface), GPT (General Purpose Timer), PWM, CAN, PORT & DIO modules.
- Interfaced analog and digital sensors with hardware using protocols like SPI, UART, USART, USB, CAN/J1939, TCP/IP, and Modbus, and wrote code for interfacing in C++.
- Implemented designs with C coding based on Object-Oriented principles on Microcontroller with VxWorks RTOS, integrating CAN/J1939 and TCP/IP protocols.
- Designed and developed USB drivers for controller boards on VxWorks, considering CAN/J1939 communication requirements.
- Produced high-quality client deliverables, including detailed test plans, test cases, test scripts, and test reports, contributing to overall client satisfaction.
- Designed, developed, and implemented Hardware-in-Loop (HIL) systems for validating and verifying mechanical, electrical, and hydraulic PLC software and associated control logic.
- Wrote scripts in Python for configuration of different embedded operating systems, including CAN/J1939 and TCP/IP settings.
- Conducted testing in accordance with guidelines on electrical solenoids, voltage level faults, mismatched wheels, missing wheels, relay faults, resistance level faults.

Environment: C++, Ubuntu, Python Control Desk, PSPICE, LABCAR, Electronic Control Unit, Hydraulic PLC, CANoe, VAS tool, Daig tool, TCP/IP, Modbus, MATLAB, CAN/J1939.

Client: Robert Bosch, India

Nov 2018 – Oct 2020

Role: Software Engineer

Responsibilities:

- The main goal of the project is to develop for different subsystems such as Engine, powertrain, and Transmission etc. of the Tractor Control Unit (OFF ROAD).
- Auto code generation of Embedded C/C++ from MATLAB Simulink and flash in to ECMs.
- Design and develop the MATLAB model for Tractor Control Unit subsystems.
- Demonstrated proficiency in Embedded C/C++ programming, delivering high-quality code that meets performance, reliability, and safety requirements.
- Worked with hybrid electric systems for tractor for different terrain.
- Development of GUIs using Control Desk for data acquisition from diff sensors.
- Writen test cases and Perform Unit testing of the Subsystems.
- Performed debugging issue in message level using CANoe and logged CAN trace.
- Hands on experience in working with 8-bit micro controllers, 16, 32 bits DSPs.
- Experience in serial protocols like CAN, SPI, I2C, LIN, MOST and experience in device drivers.
- Strong development experience in CAN, Diagnostics (UDS), OBDII, Transport protocol
- Testing the test case on HIL and performed diagnostic testing using UDS and CAN Protocol.
- Developed Software in ASPICE Compliant Process.
- Software Build of the Model and generating A2l and s19 files.

Environment: Embedded C, Python, dSpace Automation desk, Control desk, CANoe MATLAB, Simulink.

EDUCATIONAL DETAILS

Bachelors in Electronics and Communication Engineering – 2018 Graduate

Masters in Computer Engineering – 2022 Graduate